

[Chathurya V R 251801045-2025-AIDS](#) ▾**C2****Started on** Thursday, 27 August 2026, 8:25 AM**State** Finished**Completed on** Thursday, 17 September 2026, 9:41 AM**Time taken** 21 days 1 hour**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int firstZero(int arr[], int low, int high) {
4      if (low > high)
5          return low;
6
7      int mid = (low + high) / 2;
8
9      if (arr[mid] == 1) {
10         return firstZero(arr, mid + 1, high);
11     } else {
12         return firstZero(arr, low, mid - 1);
13     }
14 }
15
16 int main() {
17     int n;
18     scanf("%d", &n);
19
20     int arr[n];
21
22     for (int i = 0; i < n; i++) {
23         scanf("%d", &arr[i]);
24     }
25
26     int pos = firstZero(arr, 0, n - 1);
27
28     printf("%d\n", n - pos);
29
30     return 0;
31 }
32

```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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